ELECTENG 209

Electronics System Design

Signal Conditioning

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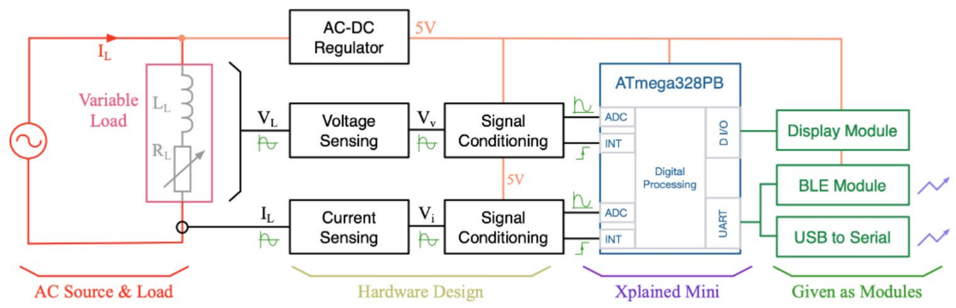
Learning Objectives

- Review of operational amplifier basics
- What are inverting and non-inverting amplifiers
- How are difference amplifiers created and why are they useful?
- Understanding practical signal conditioning circuits using single sided supplies
- What practical considerations are needed with voltage and current measurement circuits?
- How do we create RC filters and what are the design approaches in the frequency domain?
- Understanding the use and design of comparators for interrupt circuits
- Design guidelines for creating a regulated supply suited to your analogue and digital electronics.

The Energy Monitor Hardware design

- Requires Voltage Sensing, Current Sensing as discussed earlier.
- Signal conditioning
  - Level shifting, amplification and noise reduction
  - Generation of a suitable interrupt to trigger measurement from the microprocessor.
- Practical operation from a single 5V power regulator.



Practical Signal Conditioning Circuits

Level Shifting

Thevenin Equivalents

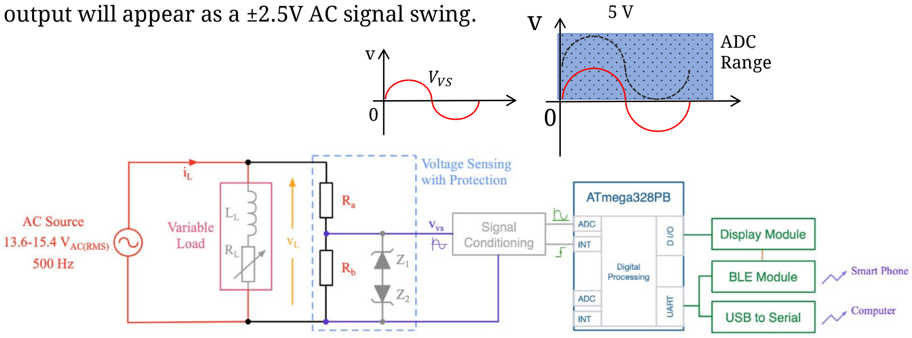
Operation with Single Sided supplies

Signal Conditioning of the Voltage and Currents

The Analogue to Digital Converter (ADC) measurement in a microprocessor only samples positive voltages (0-5V), while the AC signal has no DC component. We also normally only have one single supply to operate the op-amp off.

Therefore we need to

- Add a DC offset to the signal, then amplify and filter as required
- Use an op-amp that is operating off a single 5V supply.
- The output will appear as a ±2.5V AC signal swing.



Level Shifting and Amplifying using an Op-Amp with a Single Supply

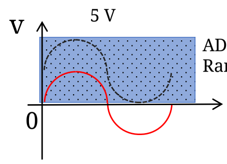
Using superposition $v^{(+)}$ can be found in terms of v_{in+} and $2.5V$ as:

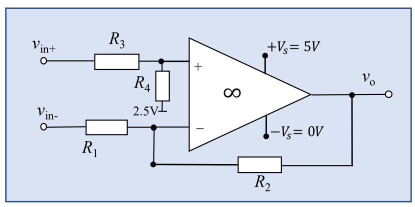
$$v^{(+)} = \frac{R_4}{(R_3+R_4)} v_{in+} + \frac{R_3}{(R_3+R_4)} 2.5 = \frac{1}{1+\frac{R_4}{R_3}} \left(\frac{R_4}{R_3} v_{in+} + 2.5 \right)$$

Show that the difference amplifier expressions are now:

$$v_o = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{1}{1+\frac{R_4}{R_3}}\right) \left(\frac{R_4}{R_3} v_{in+} + 2.5\right) - \frac{R_2}{R_1} v_{in-}$$

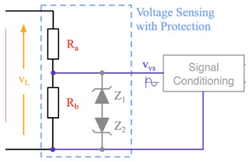
If $\frac{R_4}{R_3} = \frac{R_2}{R_1}$ then: $\quad v_o = A_{id} v_{id} + 2.5 \quad$ where $A_{id} = \frac{R_2}{R_1}$





The Thevenin Equivalent of the Voltage

- The voltage v_{vs} effective series impedance must be considered in the signal conditioning. This can be assessed using the Thevenin equivalent source.

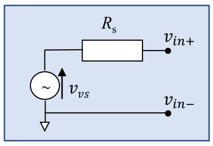


- As discussed earlier

$$v_{vs} = v_L \frac{R_b}{(R_a + R_b)}$$

This has a Thevenin equivalent series source impedance based on short circuiting v_L

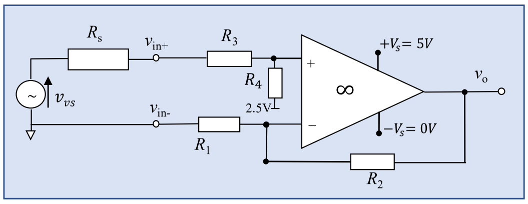
Given by: $$R_s = R_a \| R_b$$



R_s must be included in the difference amplifier design since now $R_3' = R_s + R_3$

Select:

$$R_3' = R_1$$
$$R_4 = R_2$$

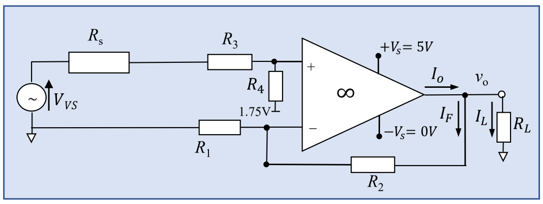


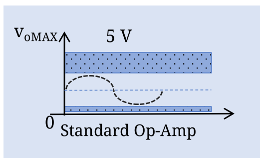
Output Voltage limitations of Op-Amps

Standard Op-Amps have outputs that cannot reach voltage supply rails, and current limits when sinking or sourcing.

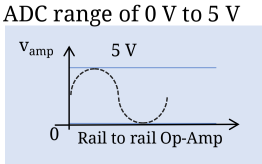
Voltage limits: The upper and lower voltage limits are L_+ and L_- respectively. Trying to output a voltage above or below these limits will cause the signal to be clipped.

- E.G., the usable output range of the LM324/358 (in your design) has $L_- = 5mV$, $L_+ = 3.5V$ (when the supply is 5 V)





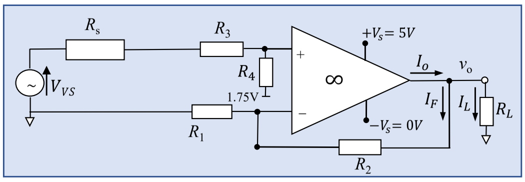
- A rail to rail Op-Amp could be used to improve accuracy
  - Output could reach close to supply rails thus allowing to utilize full ADC range of 0 V to 5 V

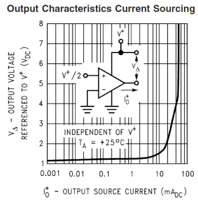


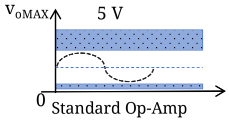
Output Current limitations of Op-Amps

Current Limits: the specified current limits (typically 20mA), include the total of the load and feedback circuit which the op-amp must supply, but the capability of the 358 to sink current at low voltage is poor.

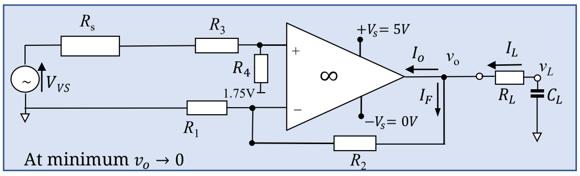
- If there is an output load R_L, when does the maximum current flow?
  - At the peak voltage (3.5V), $I_{oMAX} = I_F + I_L$. If $I_{oMAX} > I_{LIM}$ then the waveform will clip. Choose resistors in $k\Omega$

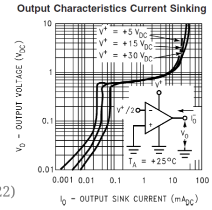






  - If there is an output filter there may be issues sinking the current into the op-amp to bring the load voltage down to < 1V that may mean we need smaller values for R_L, R_2 and R_1 to reduce the capacitor voltage in time





In general, choose:
R's < 500k
C's > 100pF

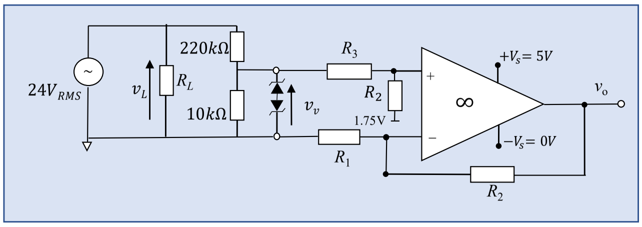
Ex: Voltage Measurement

Design a differential amplifier, that can level shift v_v measured by the voltage sensing circuit in the figure below to be within 0 to 3.5 V (within the op-amp and ADC range). Ignore the lower limitation on the Op-Amp.

- Design for a level shift of 3.5V/2 = 1.75 V

- Peak to peak level of sensed signal

$$v_{v(pk-pk)} = 24 \times 2 \times \sqrt{2} \times \frac{10k}{220k+10k} = 2.95V$$



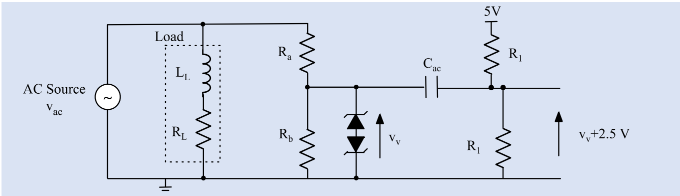
The differential amplifier gain should be no more than 1.4, but choosing $A_{id}\sim 1$ is sufficient

- Aim to achieve a 3 V signal swing at output, (0.25 V off the rail limit).
- The RMS Thevenin voltage source and resistance are: $v_{vs} = 24 \times \frac{10k}{220k+10k} = 1.04V_{RMS}$ and $R_s = \frac{220k \times 10k}{220k+10k} = 9.6k\Omega$
- If $A_{id} = \frac{R_2}{R_1} = 1$, then make $R_1 = R_2 = (R_s + R_3)$.
- We can choose $R_1 = R_2 = 100k\Omega$ and $R_3 = 90.4k\Omega \approx (22k\Omega + 68k\Omega)$ by connecting two E12 value resistors in series.

Level Shifting with a Cap

A DC bias derived through for example a voltage bias and an AC coupling capacitor can be used for level-shifting

- An example of voltage measurement is shown in figure
  - An AC coupling capacitor, C_{ac}, used to add the sensed voltage, v_v, with a 2.5V bias derived through two R_1's



C_{ac} together with Thevenin resistances of v_v and bias voltage all together form a high-pass filter
This arrangement can be used to eliminate the need for an Op-Amp

- The disadvantage is that this network cannot amplify the measured signal and will couple any high-frequency noise present, which is automatically suppressed by the op-amp due to its internal filtering.
- The capacitor can introduce some phase shift in the signal.

Ex: Current Measurement

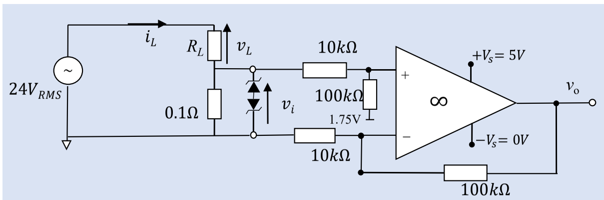
A 100 mΩ shunt resistor is used to sense the current flowing through a variable load.
The load current is in the range 0.2 A_{RMS} to 1 A_{RMS}.

Design a differential amplifier, that can amplify and level shift this signal to be within 0 to 3.5 V (op-amp range).

- Max and min peak-peak signal sensed by the shunt

$$v_{i,MAX(pk-pk)} = i_{(pk-pk)}R_s = 2\sqrt{2} \times 1A_{RMS} \times 100m\Omega = 282mV$$

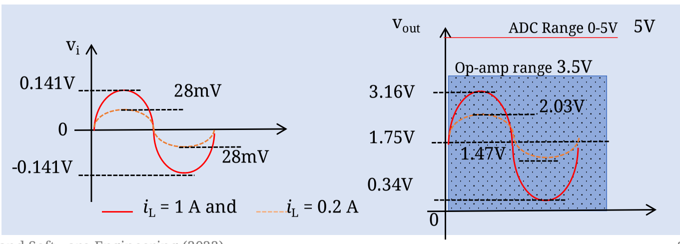
$$v_{i,MIN(pk-pk)} = i_{(pk-pk)}R_s = 2\sqrt{2} \times 0.2A_{RMS} \times 100m\Omega = 56mV$$



Again design to achieve around a 3 V signal swing at output (0.25 V off either rail)

$$A_{id} = \frac{R_2}{R_1} = \frac{3V}{0.282V} = 10.64$$

- Choosing a gain of $A_{id} \sim 10$ would satisfy this.
- Choose $R_1 = 10k\Omega$ and $R_2 = 100k\Omega$ (E12 series)
- Min and Max voltage swings are: 0.56V and 2.82V



RC Filters and the Frequency Response

A simple RC Circuit

Frequency domain Bode plots

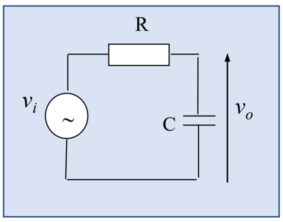
Frequency domain filtering

A Simple RC Circuit

A simple RC circuit is shown below.

- The gain of this circuit is complex and dependent on frequency because of the capacitor
- Recall from Electeng291 that using *Laplace*, the impedance of a capacitor is given by $Z_c = \frac{1}{j\omega C}$
- To find the steady state gain of the circuit in the frequency domain - analyse as a voltage divider:

$$\frac{V_o(\omega)}{V_i(\omega)} = \frac{Z_c}{Z_R + Z_c} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{1 + j\omega CR}$$



Frequency Domain Bode Plots

If the gain of a circuit is complex, then like any complex number if can be expressed in terms of its magnitude and phase.

- For $z = a + jb$

  The magnitude of this complex number is: $|z| = |a + jb| = \sqrt{a^2 + b^2}$

  The phase of this complex number is: $\angle z = \tan^{-1}\left(\frac{b}{a}\right)$

- In the case of the complex gain expression for the RC circuit, both phase and magnitude are changing with ω

Bode plots are used to view separately how the *magnitude* and *phase* vary with frequency.

However it is convenient to look at the impact using a logarithmic scale because the impact can be seen over a very wide frequency range - often over many orders of magnitudes. Our ear also has a logarithmic response which makes this assessment useful. The “bel” scale was defined as the log-base-ten of the ratio of two signal intensities related to the power or energy in the signal. As there are 10 decibels per bel the power ratio is defined as:

$$10log_{10}\frac{P_o}{P_i} \qquad \text{the power ratio in decibels (dB)}$$

Power is proportional to the square of the voltage and therefore this is re-expressed as

$$20log_{10}\frac{V_o}{V_i} \quad \text{(the voltage ratio in dB)}$$

Here the voltages are in terms of their *rms* quantities.

Frequency Domain Bode Plots

In engineering base 10 is assumed as the default when $\log H(\omega)$ is used.

- The Bode magnitude is a plot of the gain expression in dB $20\log\left|\frac{V_o(\omega)}{V_i(\omega)}\right|$ against $\log\omega$
- The Bode phase is a plot of $\tan^{-1}\left(\frac{V_o(\omega)}{V_i(\omega)}\right)$ against $\log\omega$

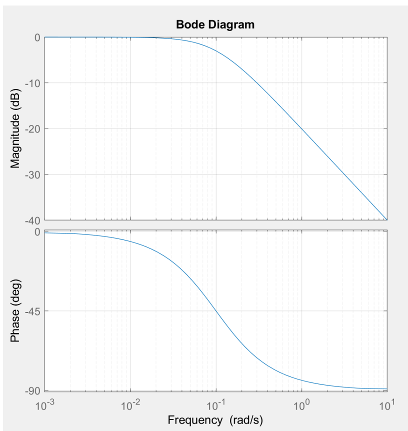
Simple log rules are useful to remember

- $\log AB = \log A + \log B$ and $\log A/B = \log A - \log B$

Creating the **Bode** plot

- Defining the time constant $T = RC$, then $G(\omega) = \frac{1}{1+j\omega T}$
- *Matlab* can be used to plot the Bode expression for any given value of T.

```
clc
T=10; % Time constant = 10s
num=1; % define numerator as 1
den=[T, 1]; % define denominator as sT+1 = jωT+1
sys = tf(num,den) % define the system transfer function
figure(1) % set up a figure to plot the system
bode(sys) % create the bode plot
grid % add a grid
```



Frequency Domain Bode Plots

Sketching approximate Bode plots

For $G(\omega) = \frac{1}{1+j\omega T}$,

Bode magnitude in dB is: $|G(\omega)|_{dB} = 20\log\left|\frac{1}{1+j\omega T}\right| = -20\log|1+j\omega T| = -\,20\log|1+j\,\omega/(1/T)|$

Bode phase is: $\angle G(\omega) = \angle\frac{1}{1+j\omega T} = -\angle(1+j\omega T) = -\tan^{-1}(\omega T/1) = -\tan^{-1}(\omega/(1/T))$

The breakpoint frequency can be defined as $\omega_b = 1/T$

- When $\omega = \omega_b$, then $|G(\omega_b)|_{dB} \cong -20\log|1+j1| = -20\log\left|\sqrt{2}\right| = -3dB$ $\quad\angle G(\omega) = -\tan^{-1}(1/1) = -45^\circ$

Asymptotes can be determined either side of ω_b

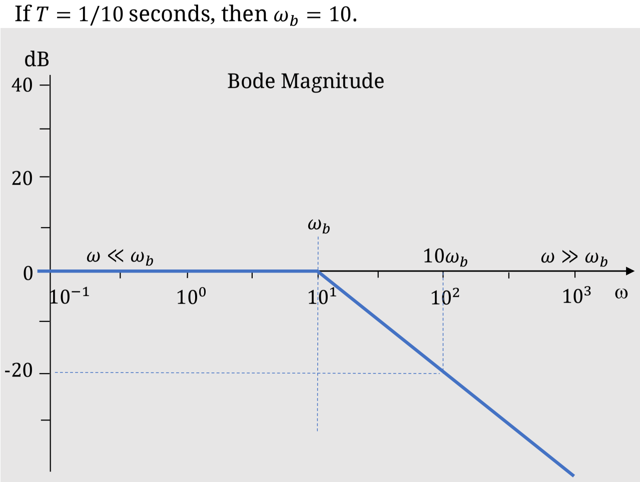
- When $\omega \ll \omega_b$, then $|G(\omega)|_{dB} = -20\log|1+j\omega/\omega_b| \cong -20\log|1| = 0dB$ $\quad\angle G(\omega \ll \omega_b) = -\tan^{-1}(0/1) = 0$
- When $\omega \gg \omega_b$, then $|G(\omega)|_{dB} = -20\log|1+j\omega/\omega_b| \cong -20\log|j\omega/\omega_b|\,dB$ $\quad\angle G(\omega \gg \omega_b) = -\tan^{-1}(\omega/\omega_b) = -90^\circ$

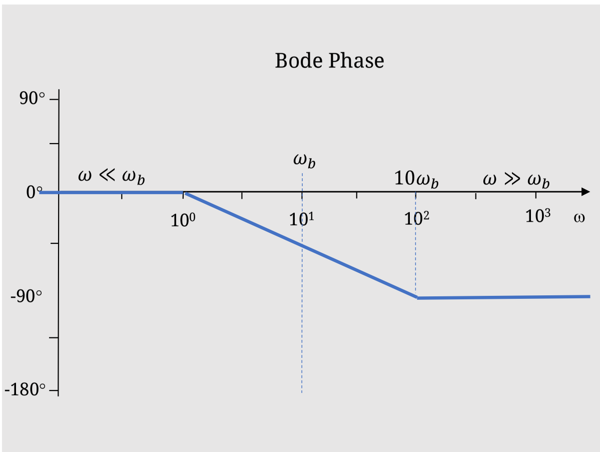
  Example: $\omega = 10\omega_b$ gives $|G(10\omega_b)|_{dB} = -20\log|j10\omega_b/\omega_b| = -20dB$, $\quad\angle G(10\omega_b) - \tan^{-1}(10) = -84.3^\circ$

  $\omega = 100\omega_b$ gives $|G(100\omega_b)|_{dB} = -20\log|j100| = -40dB$, $\quad\angle G(100\omega_b) - \tan^{-1}(100) = -89.4^\circ$

Frequency Domain Bode Plots

If $T = 1/10$ seconds, then $\omega_b = 10$.





Maximum magnitude error is 3dB at the break point frequency, while in the *phase* plot, the max error from a straight line approximation = 5.7°

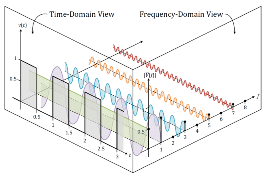
Frequency Domain Filtering

All real world signals are made up of various frequencies.

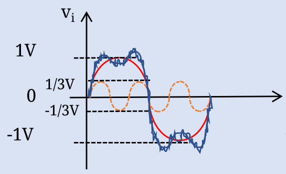
From Electeng291 notes recall that a periodic waveform in the time domain is constructed of various frequency components – and the magnitude of these can be determined from the Fourier series:

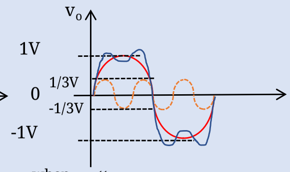
For example consider the square wave shown. When this waveform is passed through the filter:

- Frequencies at least a decade below $\omega_b = 2\pi f_b$ will have their magnitude and phase preserved.
- Frequencies above ω_b will have their magnitude reduced (attenuated) and will also be shifted in phase.
- Ex: Assume an input waveform as: $v_i = \sin\omega_f t + 0.33\sin 3\omega_f t + noise$

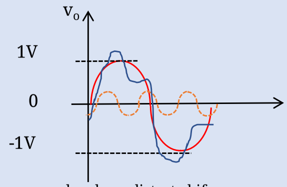


Ex: Square wave time domain v.s. frequency component amplitudes & phases





v_o when $\omega_f \ll \omega_b$
no amplitude loss and no phase distortion



v_o can be phase distorted if $\omega_f \rightarrow \omega_b$
third harmonic reduced and phase shifted

Questions?